

# Lab: Building Your Invention Roadmap

## Objective

Create an adaptive roadmap for your invention project that uses Active Inference principles: stage-gated development, expected free energy evaluation, explicit decision points, and built-in plan revision mechanisms. By the end of this lab, you will have a practical, evidence-driven project plan that balances learning with progress.

## Materials and Prerequisites

- All previous lab outputs (system map, agent profile, prediction error hunt, causal model, prototype plans, learning log, communication strategy)
- Paper or digital planning tool (a simple spreadsheet or even a hand-drawn timeline works)
- No project management background required

## Part 1: Current State Assessment (10 minutes)

Before planning where to go, you need to know where you are.

**Summarize your invention in one sentence:**

*Write your response here*

**What stage is your invention at currently?** - ☐ Idea stage — I have not validated the problem yet - ☐ Problem validated — I know the problem is real but have not tested a solution - ☐ Concept tested — I have a proof-of-concept that the mechanism works - ☐ User tested — I know users want this and can use it - ☐ Feasibility tested — I know it can be manufactured/delivered - ☐ Market ready — I am ready to launch

What are the three biggest uncertainties remaining?

| # | Uncertainty | Impact if not resolved (1-10) | Current confidence (1-10) |
|---|-------------|-------------------------------|---------------------------|
| 1 |             |                               |                           |

*Write your response here*

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|| 3 |

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**What resources do you have available?** (time per week, budget, tools, collaborators)

*Write your response here*

**Part 2: Designing Your Staged Plan (15 minutes)**

Break your project into 3-5 stages. For each stage, specify what you will learn and what decision it enables.

**Stage 1:**

| Element | Description |
|---------|-------------|
| Name    |             |

*Write your response here*

|| Primary goal (epistemic or pragmatic?) |

*Write your response here*

|| Key uncertainty to resolve |

*Write your response here*

**|| Actions to take |**

*Write your response here*

**|| Evidence that would mean "go" |**

*Write your response here*

**|| Evidence that would mean "no-go" |**

*Write your response here*

**|| Estimated time and resources |**

*Write your response here*

|

## Stage 2:

| Element | Description |
|---------|-------------|
| Name    |             |

*Write your response here*

|| Primary goal (epistemic or pragmatic?) |

*Write your response here*

|| Key uncertainty to resolve |

*Write your response here*

|| Actions to take |

*Write your response here*

|| Evidence that would mean "go" |

*Write your response here*

|| Evidence that would mean "no-go" |

*Write your response here*

|| Estimated time and resources |

*Write your response here*

|

### Stage 3:

| Element | Description |
|---------|-------------|
| Name    |             |

*Write your response here*

|| Primary goal (epistemic or pragmatic?) |

*Write your response here*

|| Key uncertainty to resolve |

*Write your response here*

|| Actions to take |

*Write your response here*

|| Evidence that would mean "go" |

*Write your response here*

|| Evidence that would mean "no-go" |

*Write your response here*

|| Estimated time and resources |



*Write your response here*

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#### Stage 4 (optional):

| Element | Description |
|---------|-------------|
| Name    |             |

*Write your response here*

|| Primary goal |

*Write your response here*

|| Key uncertainty to resolve |

*Write your response here*

|| Actions and go/no-go criteria |

*Write your response here*

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### Stage 5 (optional):

| Element | Description |
|---------|-------------|
| Name    |             |

*Write your response here*

|| Primary goal |

*Write your response here*

|| Key uncertainty to resolve |

*Write your response here*

|| Actions and go/no-go criteria |

*Write your response here*

|

### **Part 3: Building Decision Trees (10 minutes)**

For your most critical decision point (the transition from Stage 1 to Stage 2 or from Stage 2 to Stage 3), build an explicit decision tree.

**Decision point:**

*Write your response here*

**Branch A: If evidence supports the current approach:**

What do you do next?

*Write your response here*

What resources do you commit?

*Write your response here*

**Branch B: If evidence is mixed (partially supportive):**

What additional information would resolve the ambiguity?

*Write your response here*

What modified approach would you try?

*Write your response here*

**Branch C: If evidence contradicts the current approach:**

What alternative approach would you pursue?

*Write your response here*

What aspects of your work are salvageable from the current approach?

*Write your response here*

**What is the most informative observation you could make to determine which branch to follow?**

Write your response here

Part 4: Expected Free Energy Analysis (10 minutes)

Evaluate your plan using the expected free energy framework.

For Stage 1, rate each planned action on two dimensions:

| Action | Epistemic Value (how much it reduces uncertainty, 1-10) | Pragmatic Value (how much it advances toward the goal, 1-10) | Total EFE Score |
|--------|---------------------------------------------------------|--------------------------------------------------------------|-----------------|
|        |                                                         |                                                              |                 |

Write your response here

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**Based on this analysis, what should you do first?** (The action with the highest combined epistemic + pragmatic value)

*Write your response here*

**Is there an action with high epistemic value but low pragmatic value that you should do early?** (Pure learning actions)



*Write your response here*

**Is there an action with high pragmatic value but low epistemic value that you should defer?** (Actions that produce results but do not resolve uncertainty)

*Write your response here*

## Part 5: Adaptive Mechanisms (10 minutes)

Build revision mechanisms into your plan.

**Review schedule:** How often will you review and potentially revise your plan?

*Write your response here*

**Pivot criteria:** Under what conditions would you make a major change to your plan?

| Condition | Resulting Plan Change |
|-----------|-----------------------|
|           |                       |

*Write your response here*

*Write your response here*

*Write your response here*

*Write your response here*

*Write your response here*

*Write your response here*

**Planning horizon:** How far ahead can you plan with useful detail?

*Write your response here*

**What is explicitly NOT planned yet** (and should not be, because the uncertainty is too high)?

*Write your response here*

**Sunk cost check:** What would you need to see to abandon this project entirely? Write this down now, when you are not yet emotionally invested.

*Write your response here*

## **Part 6: One-Page Roadmap Summary (5 minutes)**

Synthesize your plan into a one-page summary.

**Invention:**

*Write your response here*

**Current stage:**

*Write your response here*

**Next milestone:**

*Write your response here*

**Timeline to next milestone:**

*Write your response here*

**Three biggest risks and mitigations:**

| Risk | Mitigation |
|------|------------|
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**First action (what I will do tomorrow):**

*Write your response here*

## Discussion and Debrief

1. **Epistemic vs. pragmatic balance:** Does your plan weight learning early and results later? Or does it try to produce results before critical uncertainties are resolved? How would you adjust?
2. **Decision quality:** Are your go/no-go criteria specific enough to actually make decisions? Or are they vague enough that you could justify any decision? Sharpen them.
3. **Adaptiveness:** How resilient is your plan to unexpected results? If Stage 1 produces a surprise, can your plan accommodate it without total replanning?
4. **Planning horizon humility:** Did you plan too far ahead? Is there detail in your later stages that you cannot possibly know yet?
5. **First step clarity:** Can you take the first action immediately? If not, what is blocking you, and how can you remove the blocker?

*Write your response here*